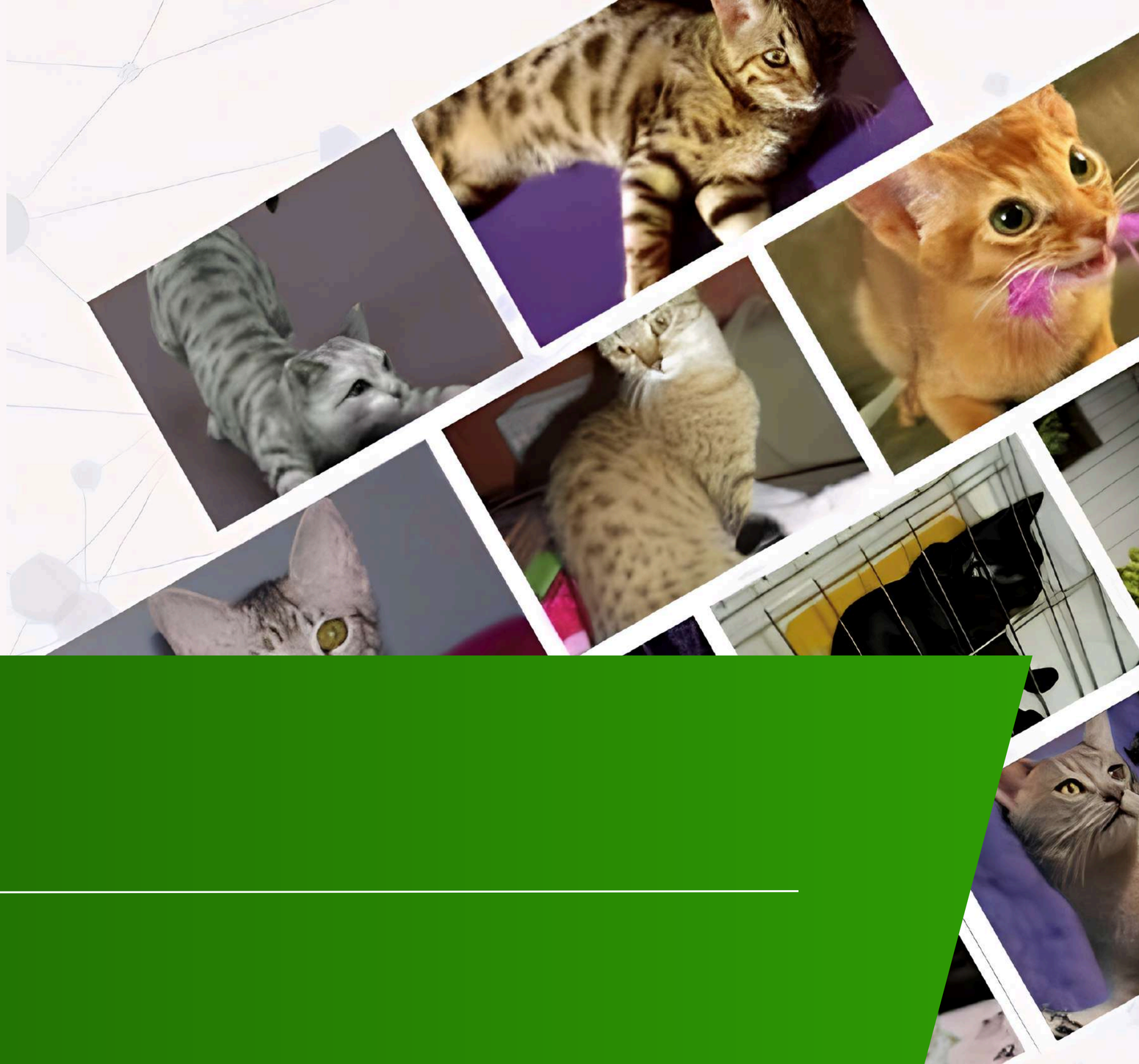
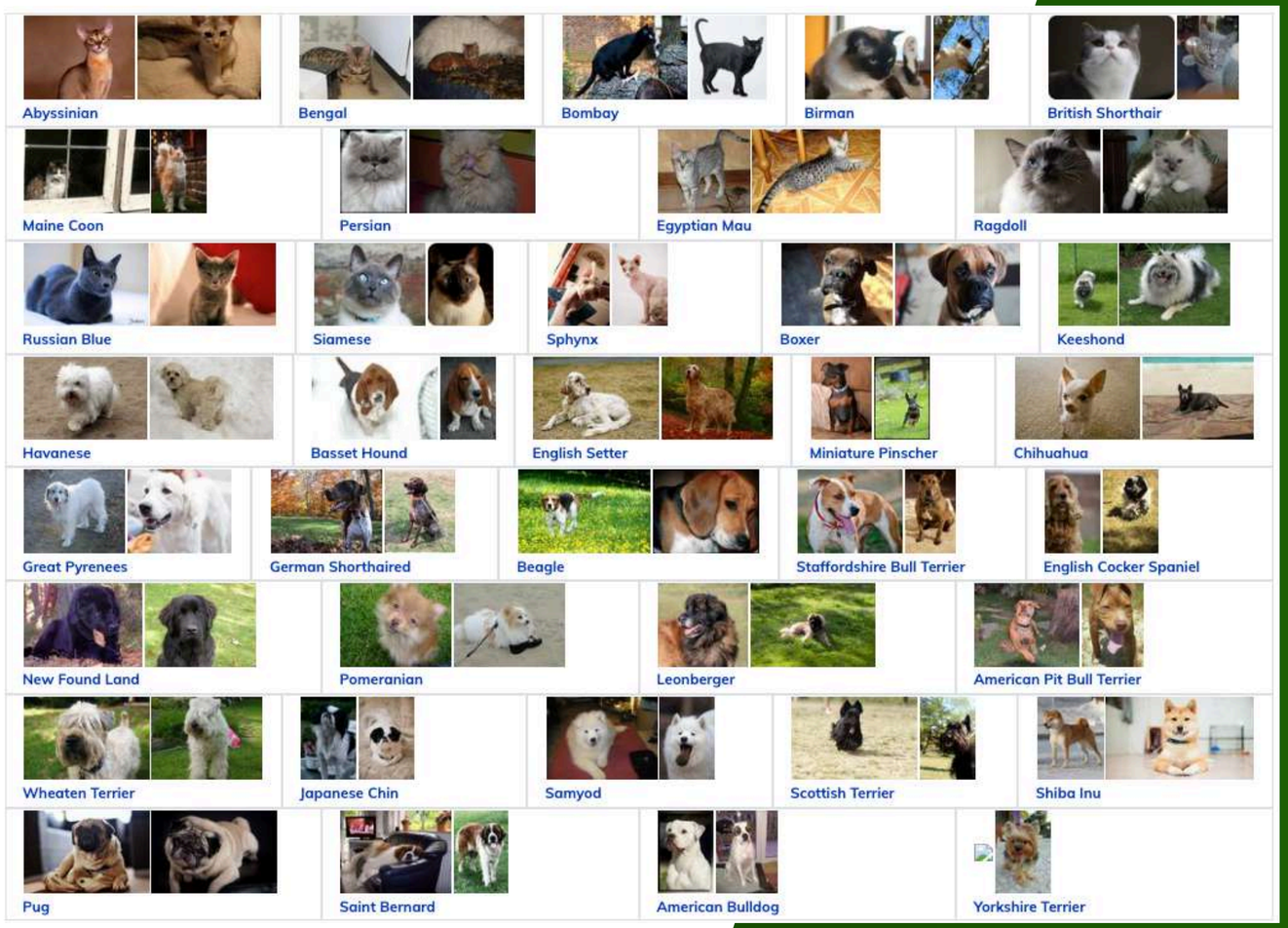
CSE428

IMAGE PROCESSING



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Reshad Ul Karim
Syeda Maliha Tabassum

DATASET



THE OXFORD-IIIT PET DATASET

Omkar M Parkhi and Andrea Vedaldi and Andrew Zisserman and C. V. Jawahar

Overview

a 37 category pet dataset with roughly 200 images for each class. The images have a large variations in scale, pose and lighting. All images have an associated ground truth annotation of breed, head ROI, and pixel level trimap segmentation.

Dataset Statistics

Breed	Count
American Bulldog	200
American Pit Bull Terrier	200
Basset Hound	200
Beagle	200
Boxer	199
Chihuahua	200
English Cocker Spaniel	196
English Setter	200
German Shorthaired	200
Great Pyrenees	200
Havanese	200
Japanese Chin	200
Keeshond	199
Leonberger	200
Miniature Pinscher	200
Newfoundland	196
Pomeranian	200
Pug	200
Saint Bernard	200
Samyoed	200
Scottish Terrier	199
Shiba Inu	200
Staffordshire Bull Terrier	189
Wheaten Terrier	200
Yorkshire Terrier	200
Total	4978

1.Dog Breeds

Breed	Count
Abyssinian	198
Bengal	200
Birman	200
Bombay	200
British Shorthair	184
Egyptian Mau	200
Main Coon	190
Persian	200
Ragdoll	200
Russian Blue	200
Siamese	199
Sphynx	200
Total	2371

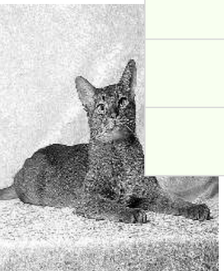
2.Cat Breeds

Family	Count
Cat	2371
Dog	4978
Total	7349

3.Total Pets

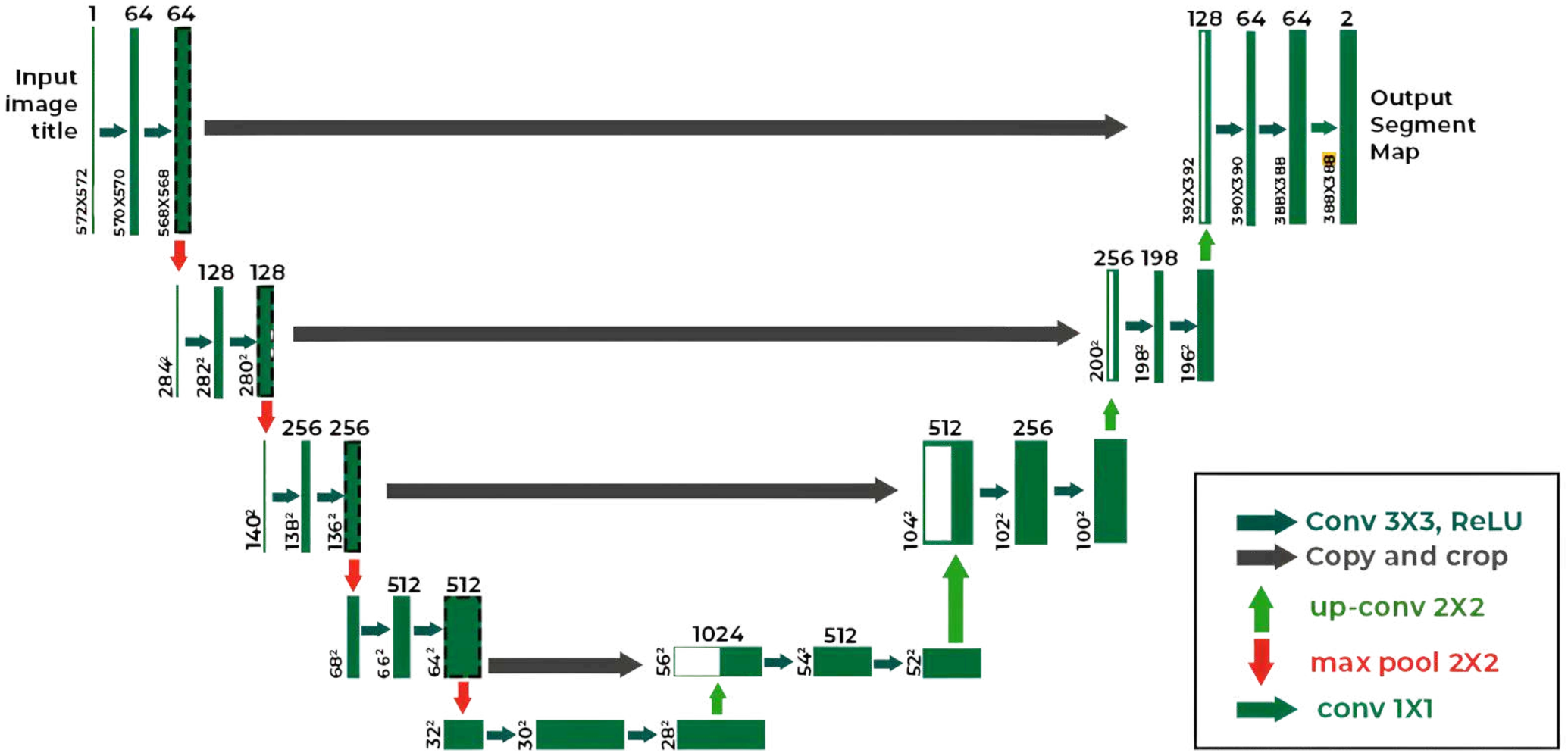
AUGMENTATION & PREPROCESSING

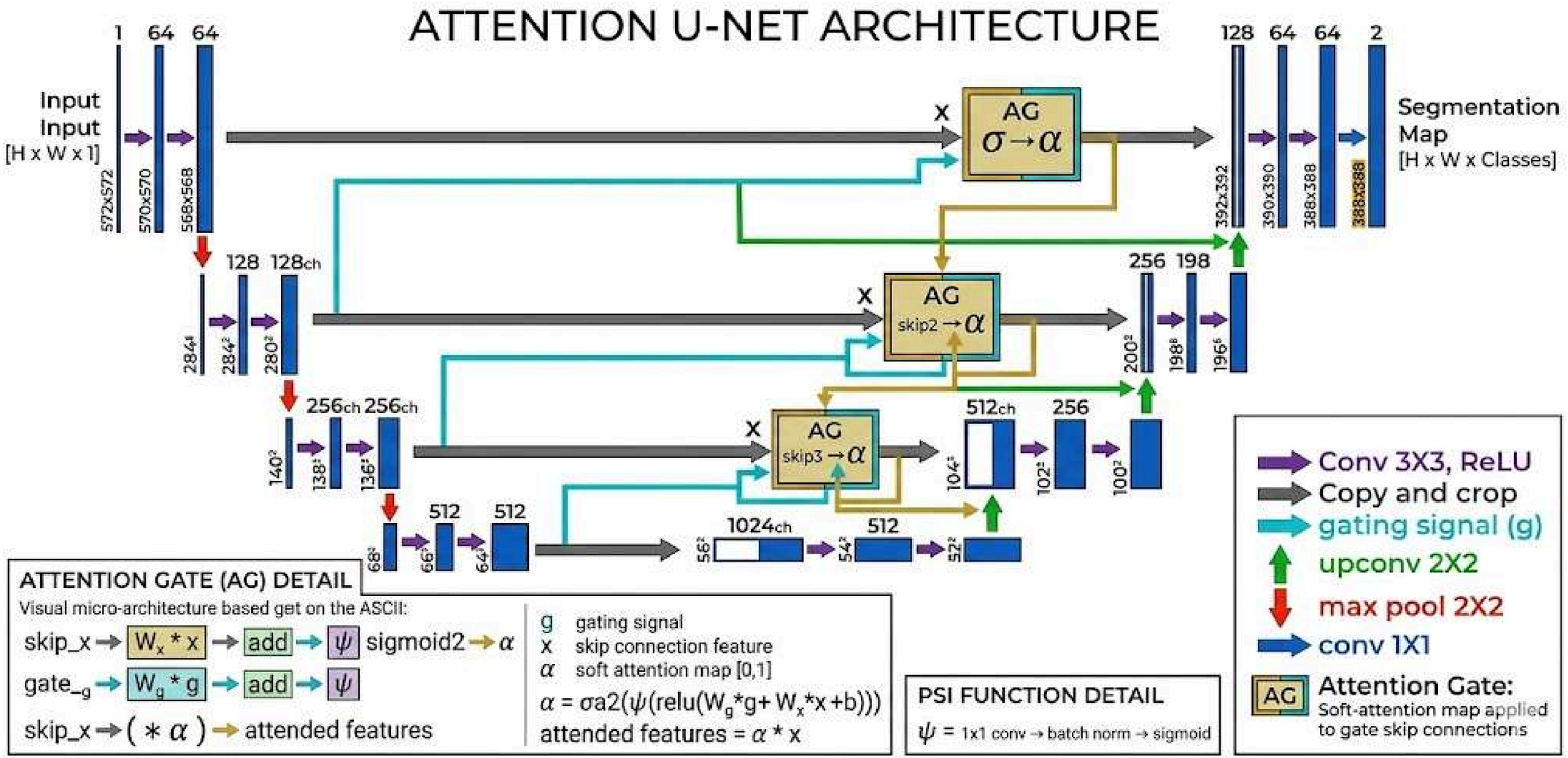
Stage	Transform / Operation	Applied To	Purpose
Input	Image loaded as HWC uint8 NumPy array	Image	Uses OpenCV/dataset-loaded image directly without PIL conversion
Input	Mask loaded as 2D NumPy array	Mask	Provides pixel-level segmentation label map
Geometric Augmentation	Horizontal Flip, p=0.5	Image + Mask	Increases left-right variation and improves generalisation
Geometric Augmentation	Rotation, limit ±15°, p=0.5	Image + Mask	Makes the model robust to small orientation changes
Geometric Augmentation	Random Resized Crop, size 256×256, scale 0.75–1.0	Image + Mask	Simulates object scale and crop variation
Photometric Augmentation	Color Jitter, p=0.6	Image only	Changes brightness, contrast, saturation, and hue to improve robustness to lighting variation
Photometric Augmentation	Grayscale Conversion, p=0.1	Image only	Encourages the model to learn shape and texture instead of relying only on colour
Noise / Blur Augmentation	Gaussian Blur, p=0.2	Image only	Simulates slight camera blur and improves robustness
Noise / Blur Augmentation	Gaussian Noise, p=0.2	Image only	Makes the model more tolerant to sensor/image noise
Regularisation	Coarse Dropout, p=0.4	Image only	Randomly removes image patches to reduce overfitting and force partial-view learning
Normalisation	ImageNet Mean and Standard Deviation	Image only	Standardises image values for stable model training
Tensor Conversion	ToTensorV2()	Image + Mask	Converts NumPy arrays into PyTorch tensors
Output	Image tensor shape: [3, 256, 256]	Image	Ready for CNN/UNet-style model input
Output	Mask tensor shape: [256, 256]	Mask	Ready for segmentation loss calculation



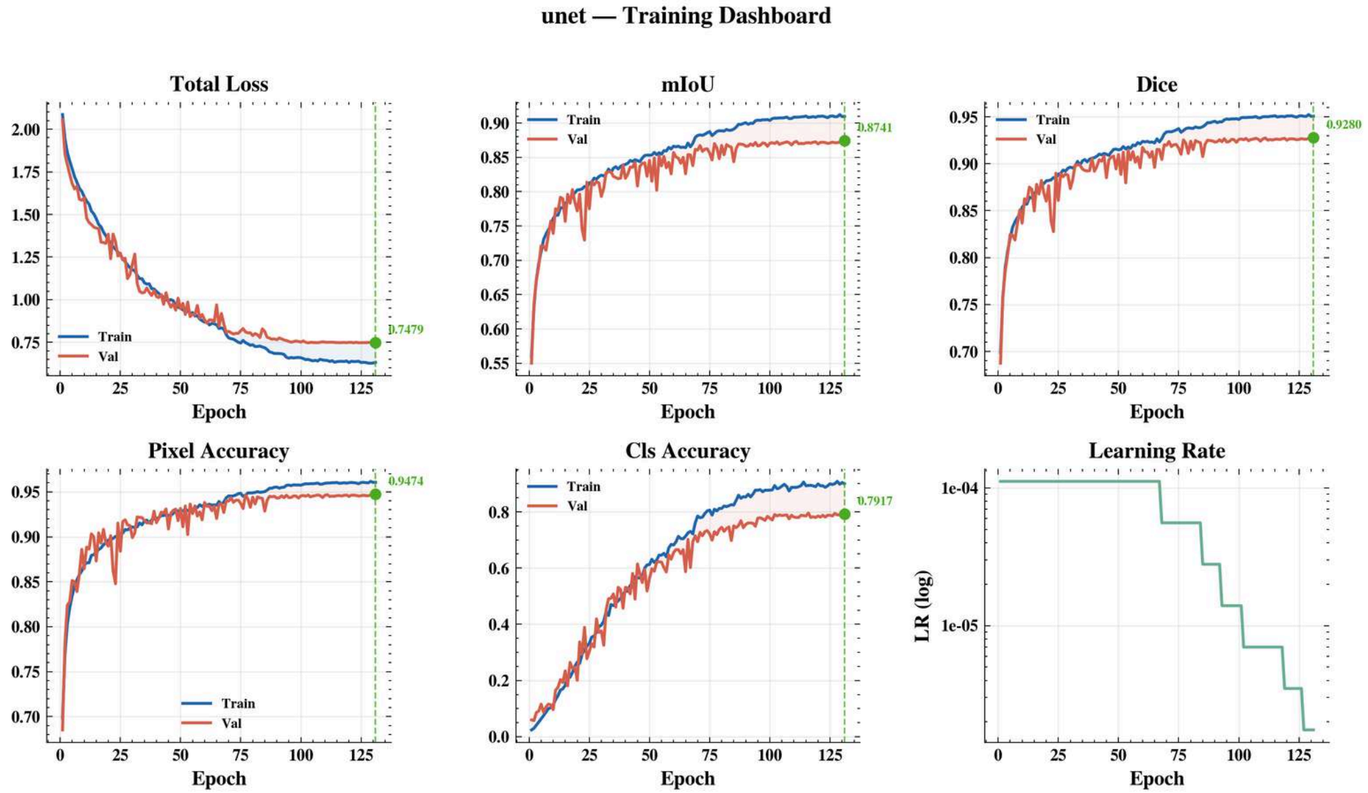
ARCHITECTURE

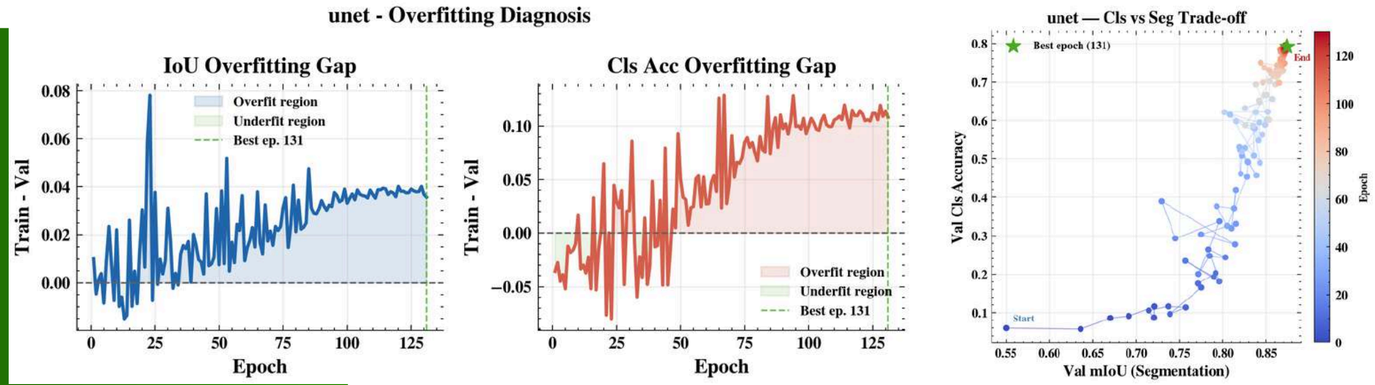
U-NET




ATTENTION U-NET ARCHITECTURE
Input
Input
[H x W x 1]
1
64
64
572x572
570x570
568x568
128
128ch
284²
282²
280²
256ch
256ch
140²
138²
136²
512
512
68²
66²
64²
1024ch
56²
512
54²
52²
AG
σ→α
AG
skip2→α
AG
skip3→α
x
x
x
256
198
200²
198²
196²
512ch
256
104²
102²
100²
128
64
64
2
392x392
390x390
388x388
388x388
Segmentation
Map
[H x W x Classes]
Conv 3X3, ReLU
Copy and crop
gating signal (g)
upconv 2X2
max pool 2X2
conv 1X1
AG
Attention Gate:
Soft-attention map applied
to gate skip connections
ATTENTION GATE (AG) DETAIL
Visual micro-architecture based get on the ASCII:
skip_x → Wx * x → add → ψ sigmoid2 → α
gate_g → Wg * g → add → ψ
skip_x → (* α) → attended features
g gating signal
x skip connection feature
α soft attention map [0,1]
α = σa2(ψ(relu(Wg*g+ Wx*x +b)))
attended features = α * x
PSI FUNCTION DETAIL
ψ = 1x1 conv → batch norm → sigmoid

UNET

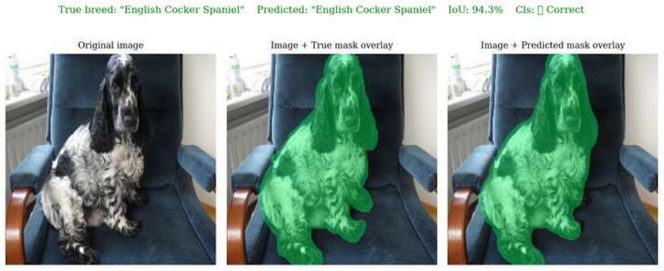




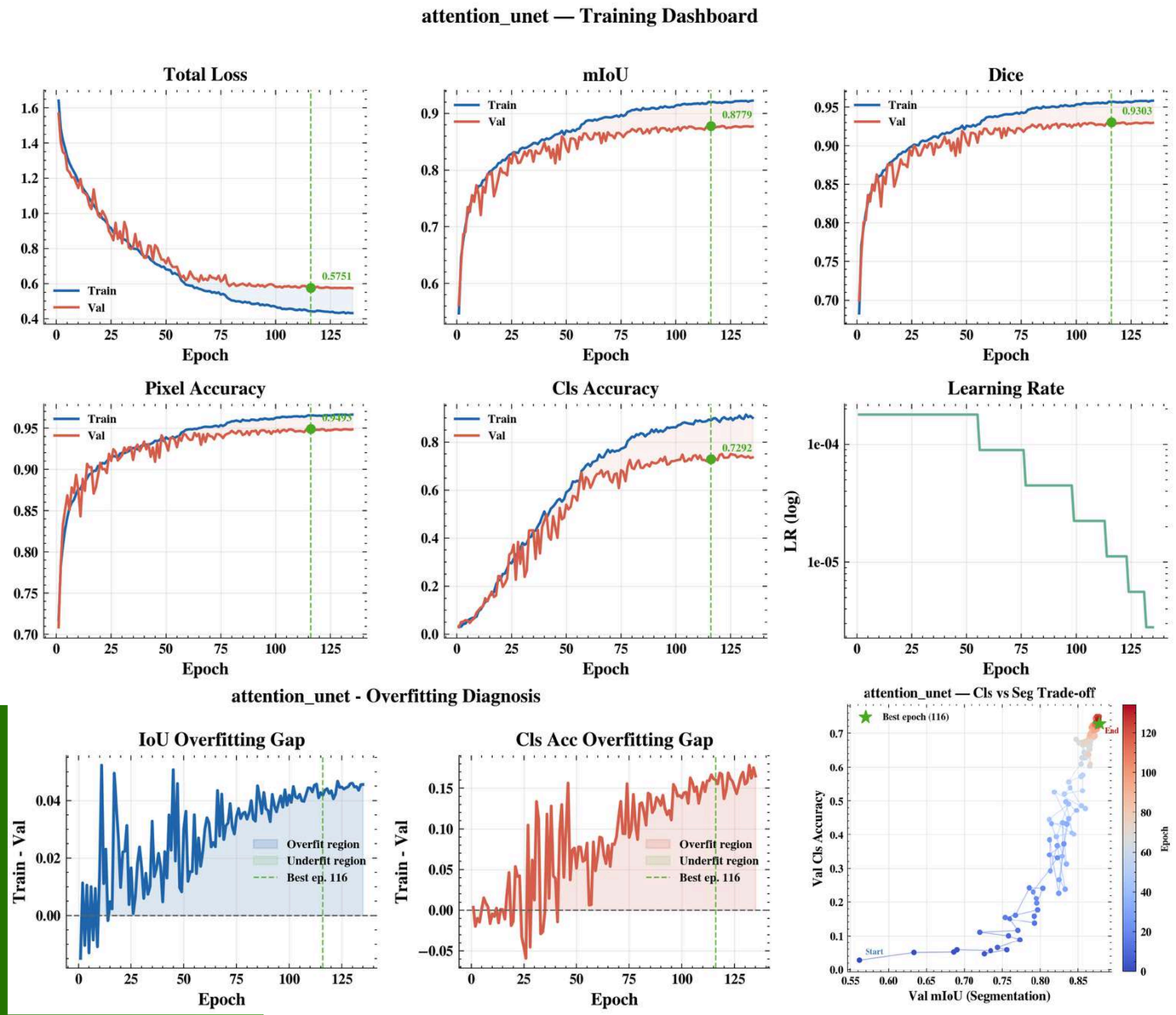
HYPERPARAMETER TUNED

Category	Metric / Parameter	Value
Optimization Summary	Objective Metric	val_iou
	Best Val IoU	**0.761673**
	Best Trial Number	13
	Total Trials	30
	Epochs per Trial	8
Best Hyperparameters	Learning Rate	0.0001117456
	CLS Lambda	0.3599288613
	Weight Decay	0.0004078854
	Dropout Rate	0.5320063444
	CLS WD Multiplier	50
	Scheduler Patience	7

Category	Metric	Value
Segmentation	mIoU	0.884428879591331
	Dice	0.934353059186702
	Pixel Accuracy	0.951269412292621
Classification	Accuracy	0.769692014172799
	Precision	0.776506558544901
	Recall	0.768972762415614
	F-1. Score	0.767437308298722



ATTENTION U-NET

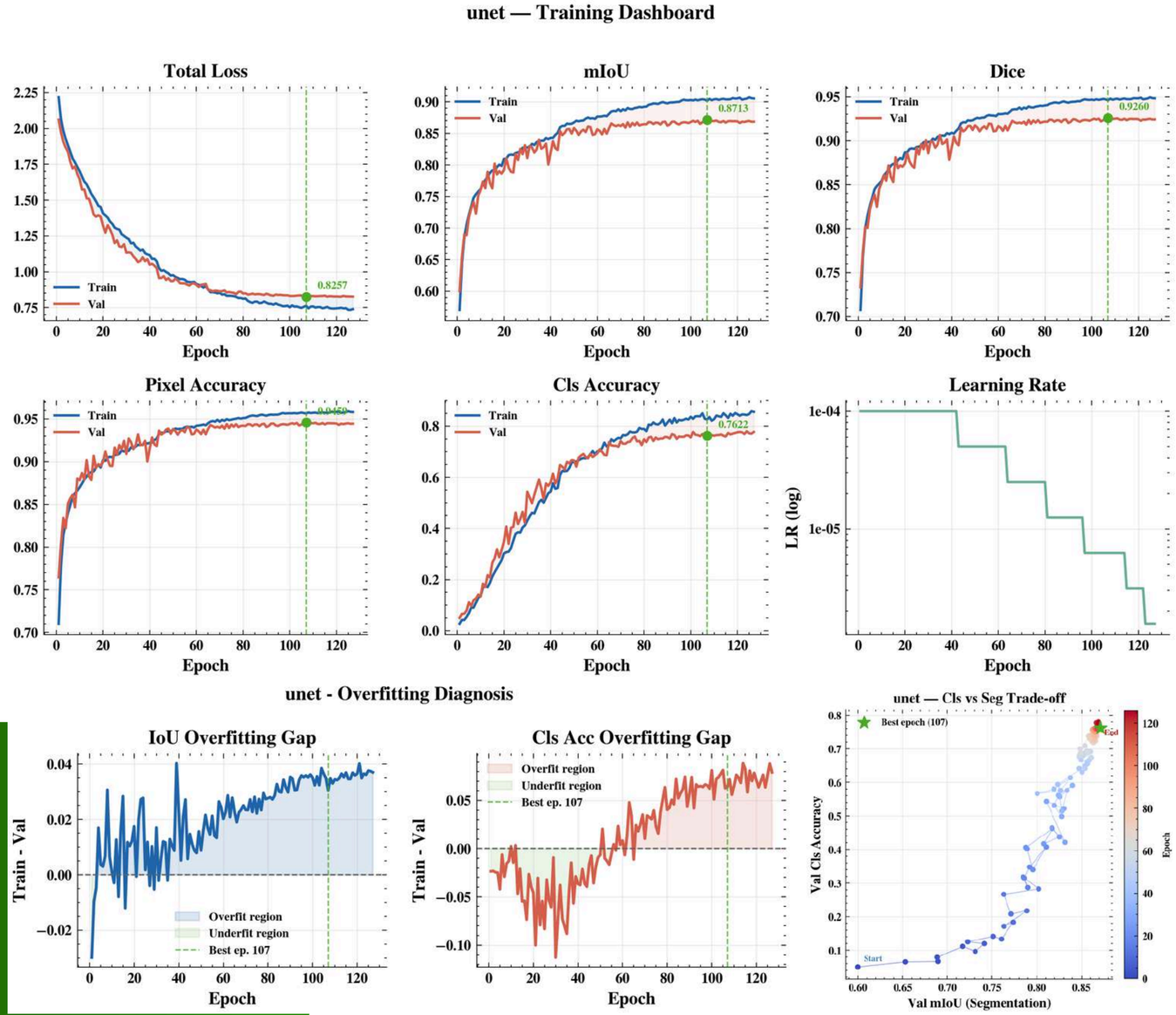


HYPERPARAMETER TUNED

Category	Metric / Parameter	Value
Performance Summary	Objective Metric	**val_iou**
	Best Validation IoU	**0.775684**
	Best Trial	23
	Total Trials	30
	Epochs per Trial	8
Optimal Hyperparameters	Learning Rate	0.00017902
	CLS Lambda	0.23606962
	Weight Decay	0.00027469
	Dropout Rate	0.55918269
	CLS WD Multiplier	20
	Scheduler Patience	10

Category	Metric	Value
Segmentation	mIoU	0.889391325719555
	Dice	0.937484473744693
	Pixel Accuracy	0.953580555312546
Classification	Accuracy	0.74897792313982
	Precision	0.758195313244439
	Recall	0.748350567344792
	F1-Score	0.74777114106487

BI-FPN_UNET ADAPTATION FROM EFFICIENTDET

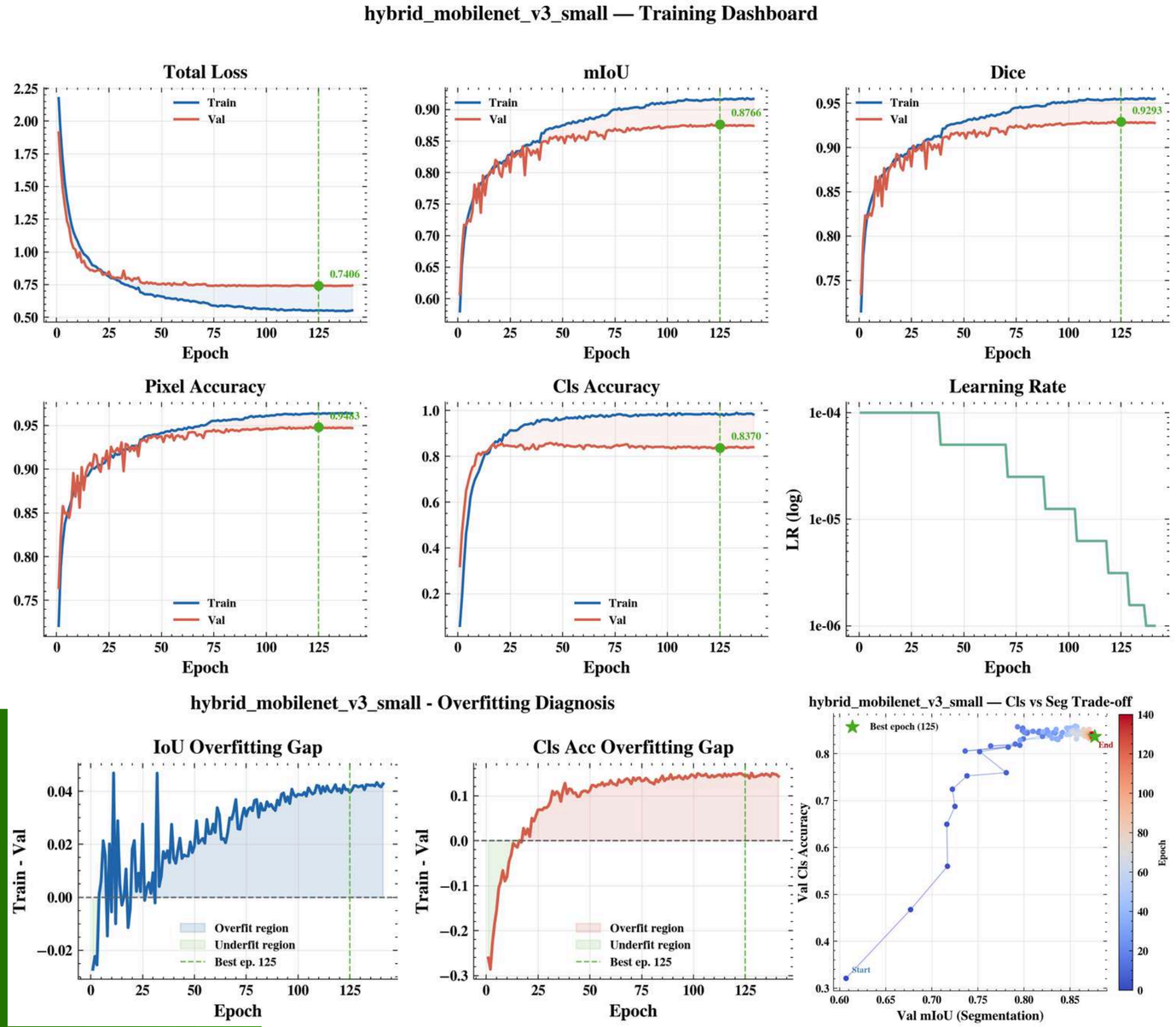


Adapted BiFPN (Bidirectional Feature Pyramid Network) from EfficientDet and integrated in the decoder side of the generic UNET architecture to improve classification accuracy and efficiency.

Category	Metric	Value
Segmentation	mIoU	0.883344389932224
	Dice	0.933676798718732
	Pixel Accuracy	0.950609192063015
Classification	Accuracy	0.749523030798582
	Precision	0.7577678103123
	Recall	0.749050118032345
	F1-Score	0.747535064190547

*hyperparameters are based on empirically tuned unet run

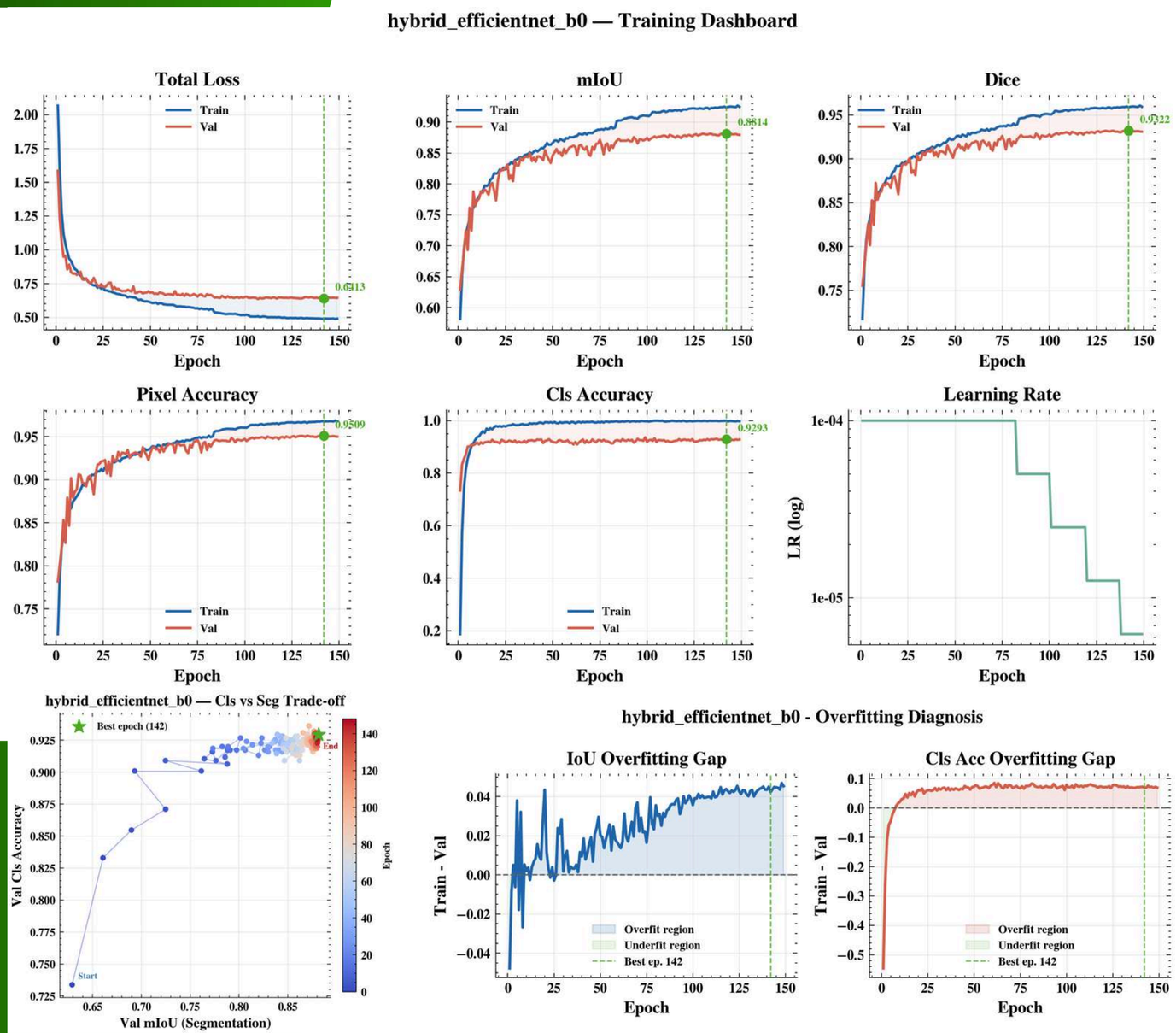
UNET + MOBILENET SMALL CLASSIFIER HEAD



Category	Metric	Value
Segmentation	mIoU	0.889115103154263
	Dice	0.937084490231469
	Pixel Accuracy	0.953413964602961
Classification	Accuracy	0.825565549195966
	Precision	0.82825205767962
	Recall	0.825419086544998
	F1-Score	0.82346807201041

*hyperparameters are based on empirically tuned unet run

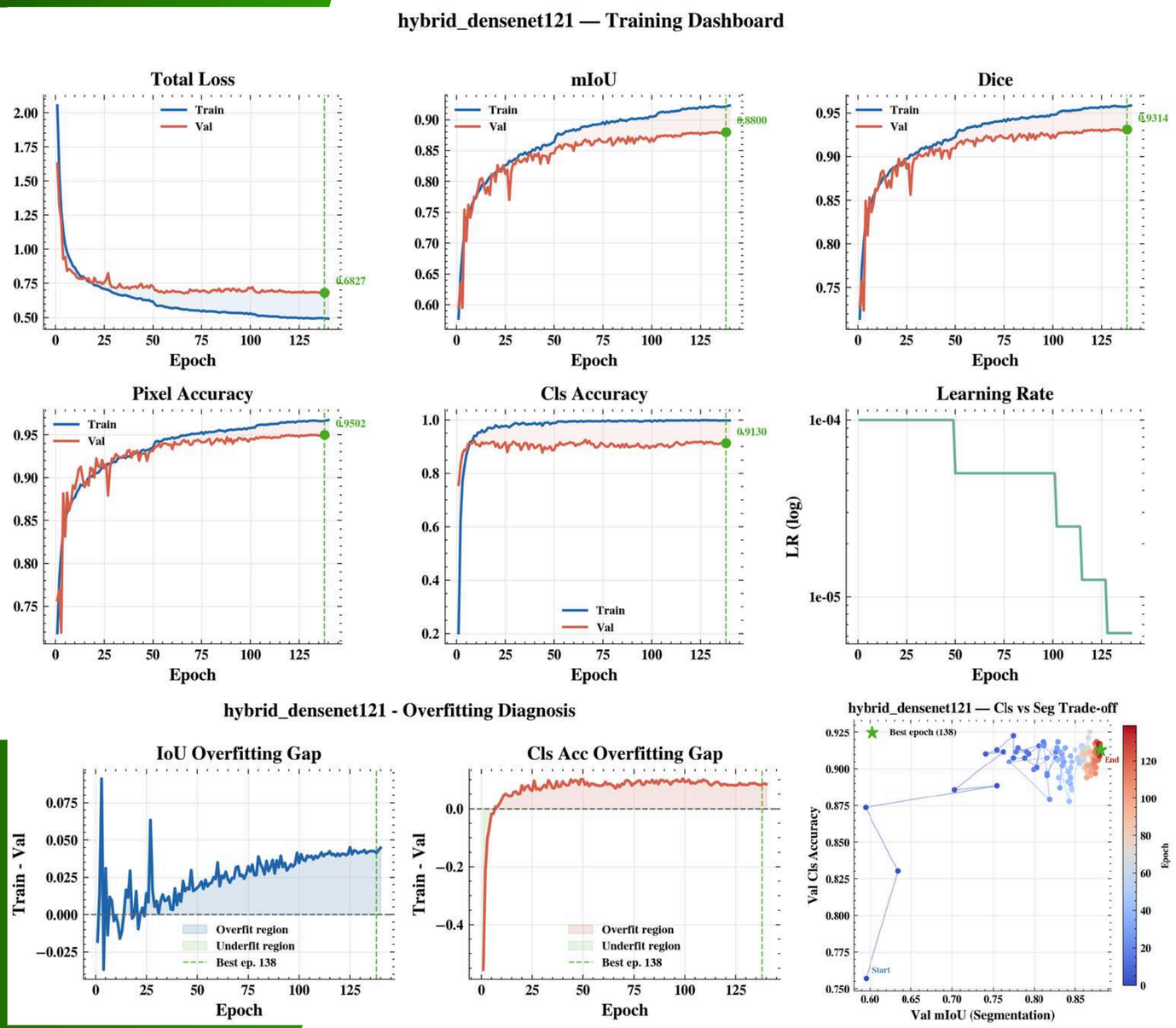
UNET + EFFICIENTNET_B0 CLASSIFIER HEAD



Category	Metric	Value
Segmentation	mIoU	0.89210599961192
	Dice	0.938806806861749
	Pixel Accuracy	0.954810204105358
Classification	Accuracy	0.913872989915508
	Precision	0.914578490814849
	Recall	0.913081913584222
	F1-Score	0.912173020569502

*hyperparameters are based on empirically tuned unet run

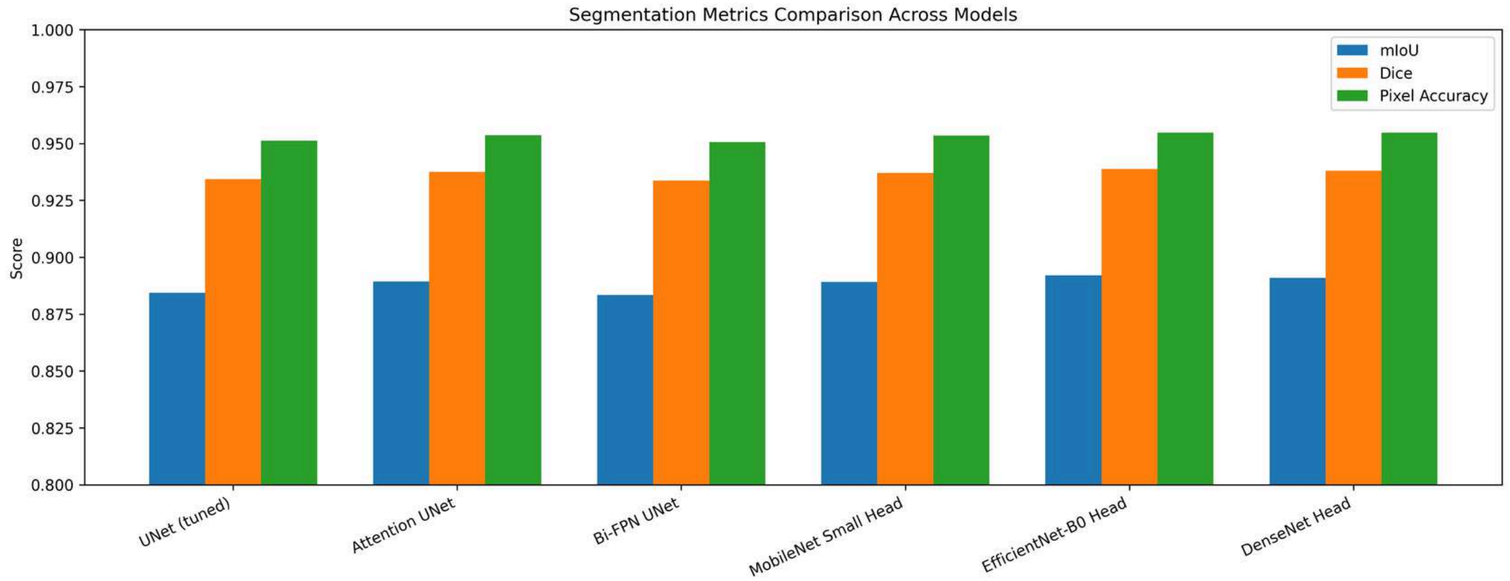
UNET + DENSENET CLASSIFIER HEAD

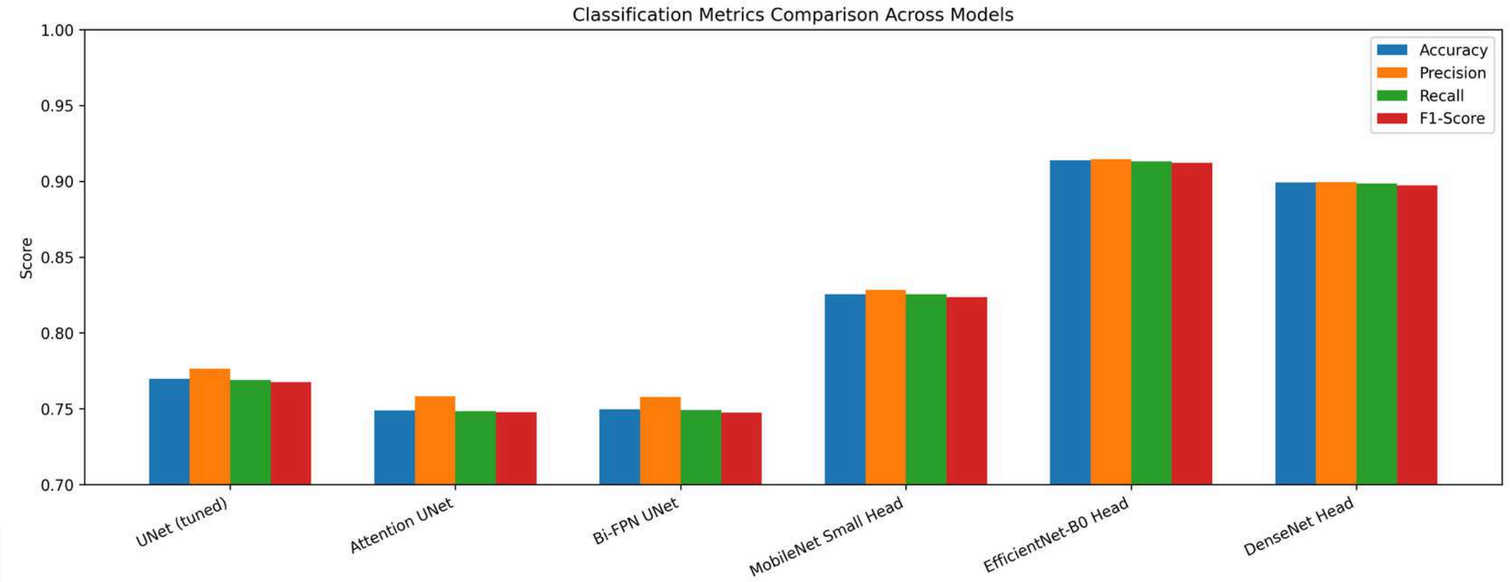


Category	Metric	Value
Segmentation	mIoU	0.890984232706614
	Dice	0.937906629672111
	Pixel Accuracy	0.954680518954274
Classification	Accuracy	0.899155083128918
	Precision	0.899466423270173
	Recall	0.898378519526109
	F1-Score	0.897302600129558

*hyperparameters are based on empirically tuned unet run

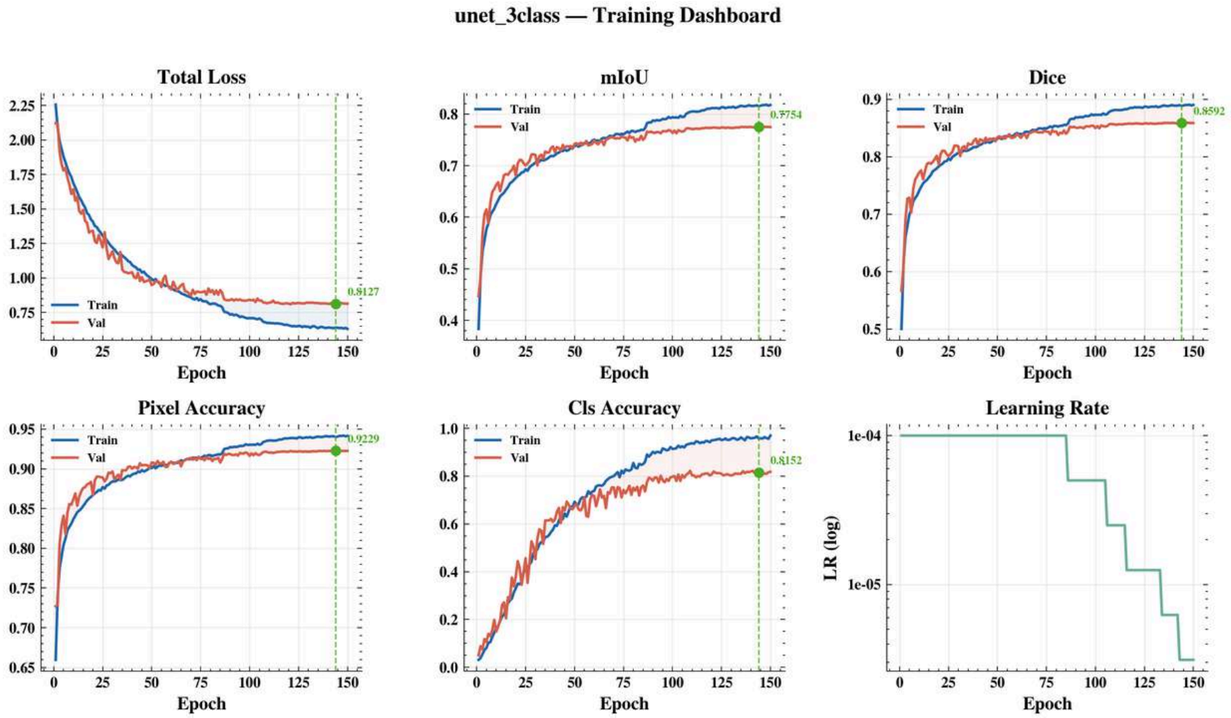
OVERALL COMPARISON

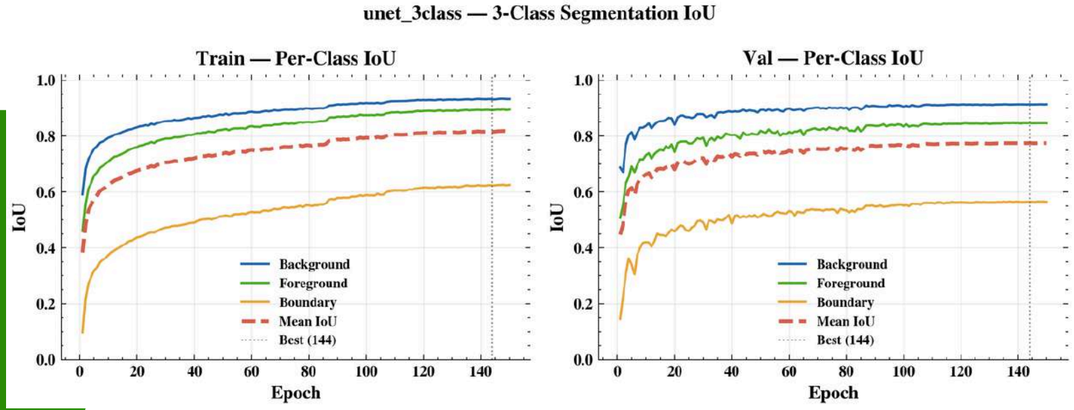




TREATING SEGMENTATION AS A 3 CLASS PROBLEM

- BACKGROUND
- FOREGROUND
- BOUNDARY





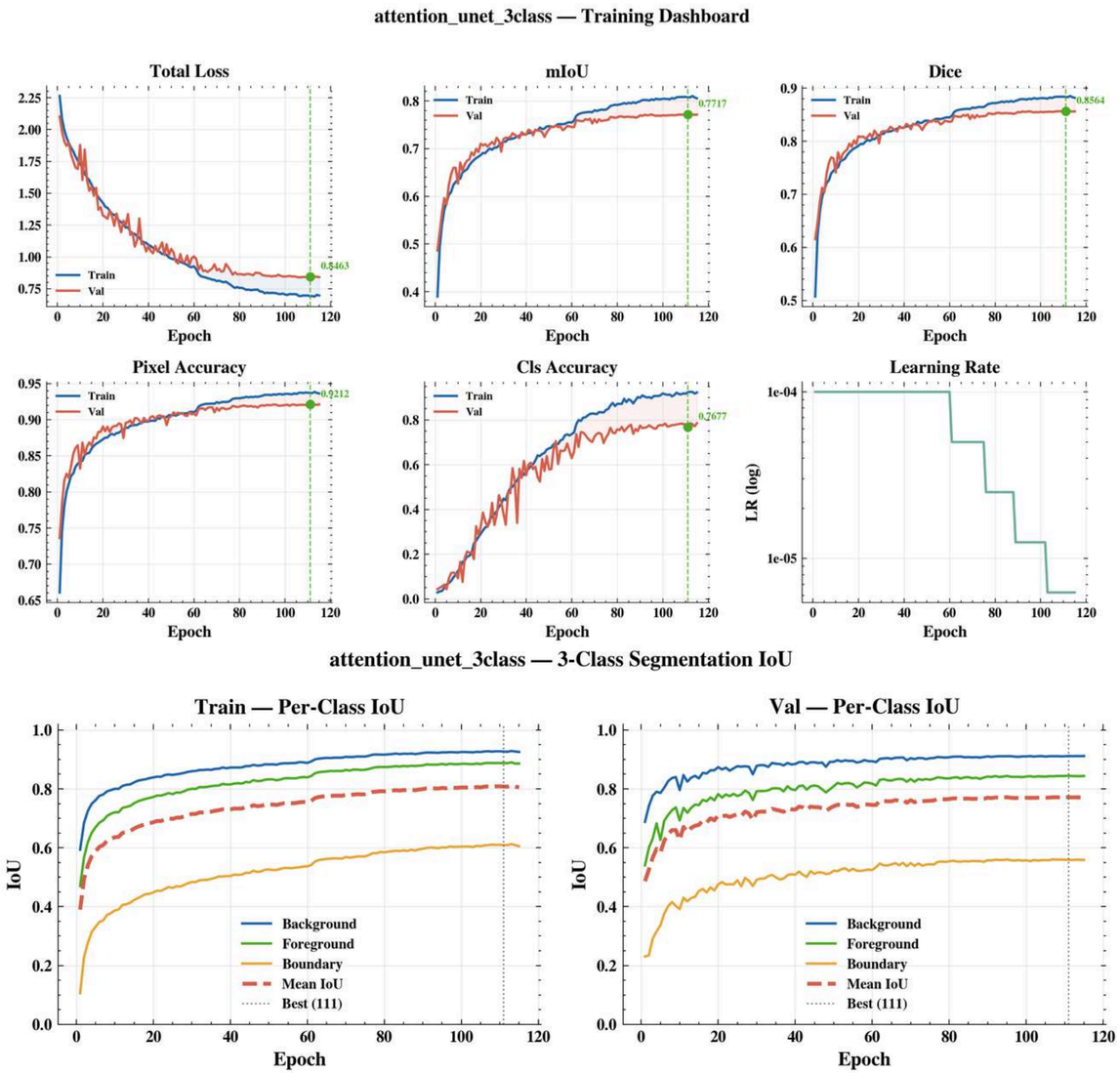
UNET

Category	Metric	Value
Segmentation	mIoU	0.783981729763141
	IoU (Background)	0.915139409770136
	IoU (Foreground)	0.848055067269698
	IoU (Boundary)	0.58875071224959
	Mean Dice	0.86617309252421
	Pixel Accuracy	0.924982498003089
Classification	Accuracy	0.808394657944944
	Precision	0.811732592897685
	Recall	0.807530711559438
	F1-Score	0.806306884697763

*hyperparameters are based on empirically tuned unet run

TREATING SEGMENTATION AS A 3 CLASS PROBLEM

- BACKGROUND
- FOREGROUND
- BOUNDARY



Category	Metric	Value
Segmentation	miou	0.77935370120449
	iou_bg	0.91191782173903
	iou_fg	0.84263810696809
	iou_boundary	0.58350517490635
	mean_dice	0.86282533389934
	pixel_acc	0.92265707409900
Classification	accuracy	0.76941946034341
	precision	0.77922150932001
	recall	0.76859063527263
	f1	0.76741313455435

*hyperparameters are based on empirically tuned unet run

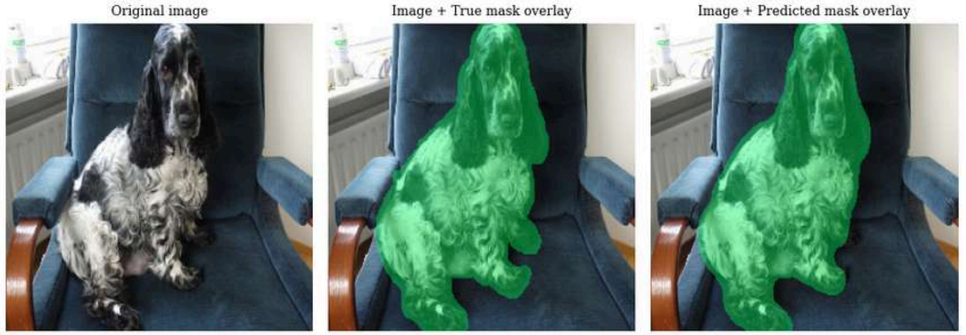
ATTENTION UNET

INFERENCE SAMPLE

True breed: "Chihuahua" Predicted: "Chihuahua" IoU: 85.1% Cls: □ Correct



True breed: "English Cocker Spaniel" Predicted: "English Cocker Spaniel" IoU: 94.3% Cls: □ Correct



True breed: "Bengal" Predicted: "Bengal" IoU: 74.8% Cls: □ Correct

